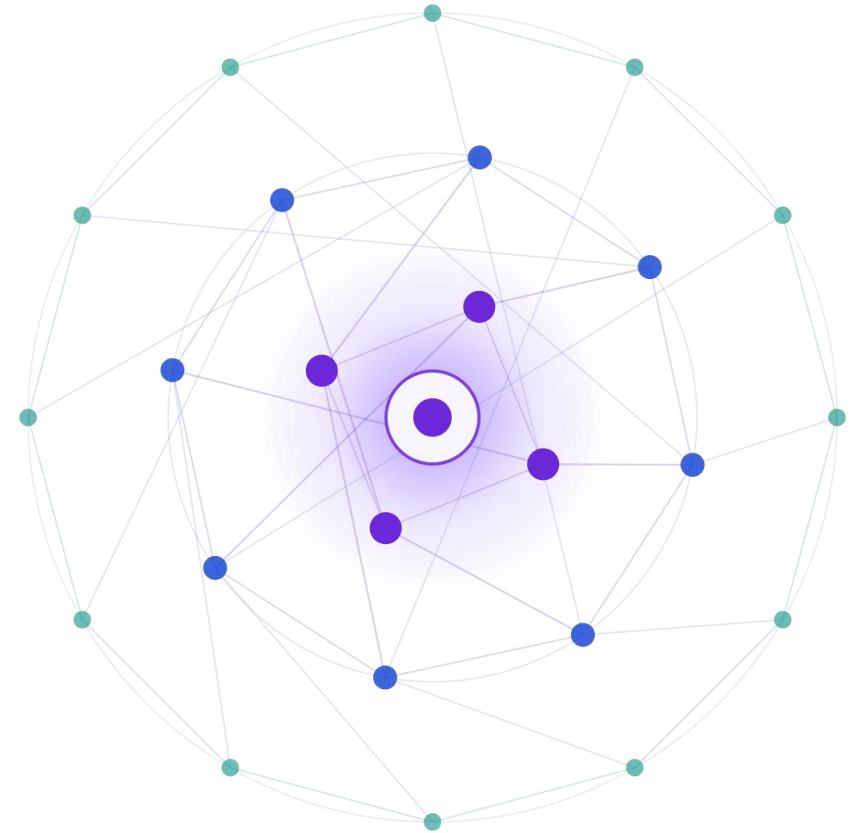


Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - 02
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- Submission Github link -
github.com/pragadeeswarrs/smart_guided_troubleshooting_engine



Theme

THEME ID 02

Smart Guided Troubleshooting Engine

Turn a vague, natural-language smartphone complaint into a structured, actionable recovery plan with native Samsung Bixby deeplinks.

Vague complaint

"My Wi-Fi keeps disconnecting and battery is draining fast"



Structured plan

Grouped action cards, sorted
auto → manual → critical



One-tap fix

Each step opens the exact
Settings screen via a deeplink

Existing Solutions & Gaps

~15 min

per scenario

agents manually translate a vague
complaint into troubleshooting steps

× millions of calls

Existing solution: human agents

Customer-service agents read the complaint, guess the root cause and hand-type steps. Slow, costly, and steps vary from agent to agent.

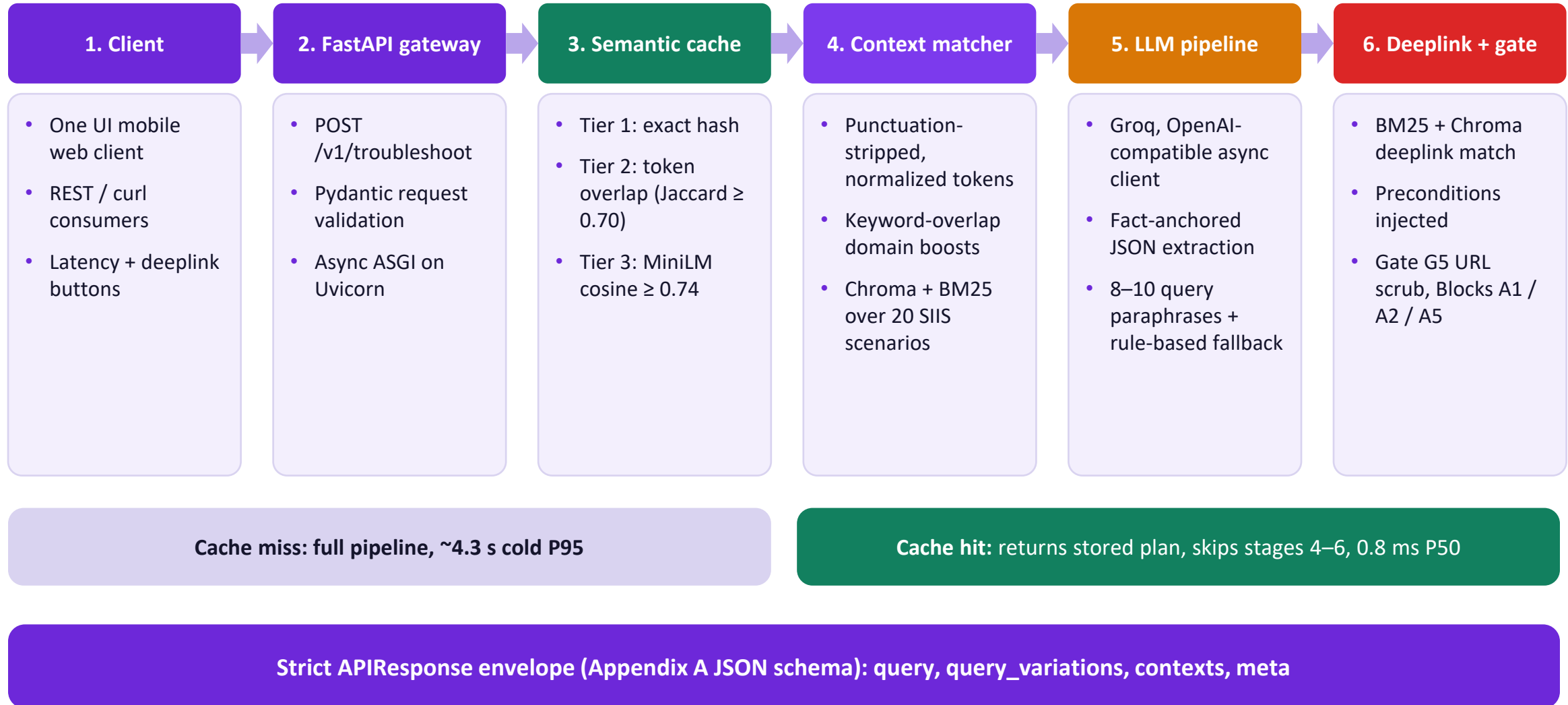
Existing solution: self-service menus

Users still have to hunt through nested Settings menus themselves to find the right screen and toggle.

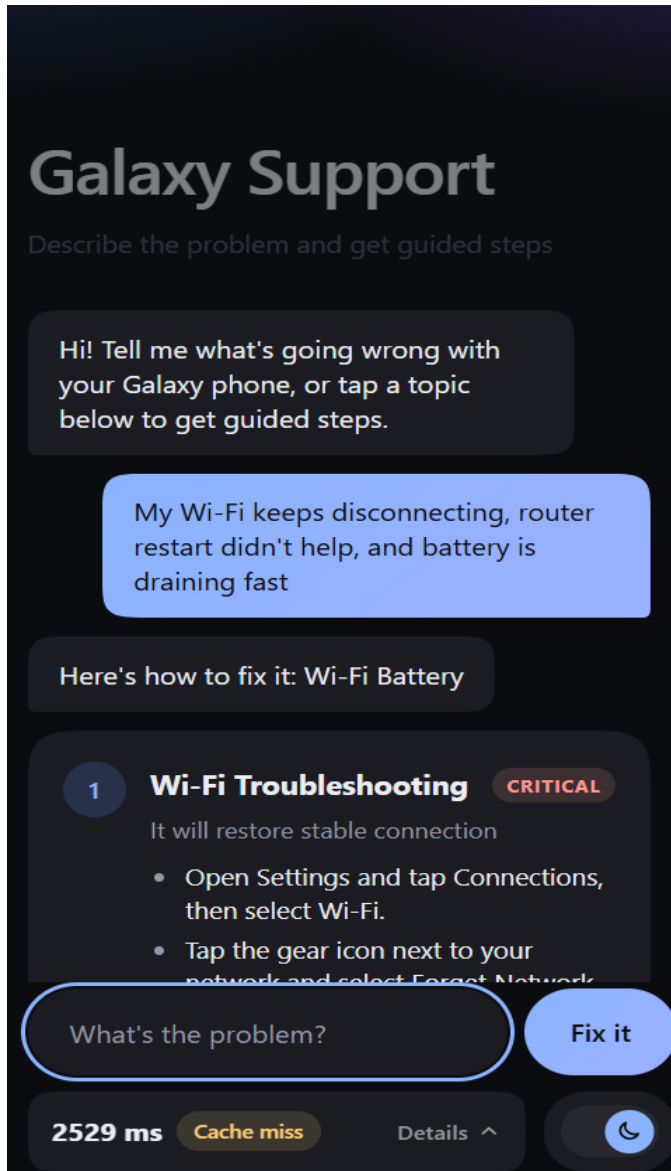
The gap

No instant path from “it’s broken” to the exact Settings screen, and no consistent, precondition-checked fix at scale.

Our Solutions & Architecture Diagram



Demo & Product Walkthrough



1

Grouped action cards

One card per detected problem, so a composite complaint (Wi-Fi + battery) yields a Battery card and a Wi-Fi card in one plan.

2

Tri-category actions

Steps are sorted auto → manual → critical with colour-coded badges; critical steps carry a distinct warning.

3

Actionable deeplinks

A single tap opens the exact Settings screen, e.g. "Wi-Fi Settings" → `bixby://com.samsung.android.settings.connections.wifi`. Manual steps show no button.

4

Live diagnostics

Latency badge flashes green on cache hit and amber on miss; a drawer shows model and request cost.

Tools and tech stack used

Python 3.10+

Core language for the engine

FastAPI + Uvicorn

Async REST API, Swagger docs, ASGI server

Pydantic v2

Strict Appendix A schema validation

Groq API

Llama 3 / gpt-oss-20b via OpenAI-compatible SDK

SentenceTransformers

all-MiniLM-L6-v2, 384-dim embeddings

ChromaDB

Local vector store for deeplink + intent search

Rank-BM25

Lexical retrieval fused with vector search

DiskCache

Persistent multi-tier semantic cache

Frontend: vanilla HTML5 / CSS3 / JS One UI client | Evaluation: httpx benchmark suite (20 scenarios)

Impact & Use case

Vague complaint

“My Wi-Fi keeps disconnecting and battery drains fast”



Grounded plan

Matched to Samsung SIIS scenarios; steps sorted and validated



One-click fix

Deeplink buttons open the exact Settings screen

~15 min → ~4 s

Agent translation replaced by a cold-start plan (P95 4.3 s); repeat queries return in ~1 ms.

Self-service for users

No menu hunting: every actionable step is one tap away.

Consistent and safe

Same contract-valid plan every time; critical actions flagged, preconditions attached, zero URL leaks.

Innovation highlights, results and limitations

Innovation highlights

- Keyword-overlap context finder: normalized, punctuation-stripped tokens + domain boosts feed dense + BM25 retrieval
- Strict JSON derivation rules: goal regex, 2–3 word titles, “It will...” descriptions, 8–10 variations
- Multi-tier semantic cache with sub-2 ms fast path
- Multi-intent complaints resolved in one plan

Results

- Cache hit: 0.8 ms P50, 1.7 ms P95 (target ≤ 300 ms)
- Cold start: 4,279 ms P95 (target $\leq 8,000$ ms)
- 20 / 20 scenarios covered
- 100% Appendix A schema compliance
- 0 URL leaks; 100% cache and paraphrase hit rate

Limitations

- Knowledge base covers 20 SIIS scenarios
- Cold-start latency (~4 s) is LLM-bound
- Rule-based fallback is less rich than LLM output
- Preconditions are declared, not read from a live device
- Cache thresholds (0.70 / 0.74) are tuned on a small set

What's next

1

Wider knowledge base

Grow beyond 20 SIIS scenarios and expand the Bixby deeplink catalog.

2

Live device state

Evaluate preconditions (wifi_enabled, brightness) against the real device before showing a step.

3

Multilingual complaints

Accept and answer in regional languages, starting with Indian languages.

4

Agent-assist and feedback

Plug into agent tooling and learn from “did it fix it?” signals to rerank steps.

Brownie points slide (differentiation)

Model-agnostic LLM layer

Built on an OpenAI-compatible client: Groq (Llama 3 / gpt-oss-20b) today, and a Gemini or Mistral endpoint only needs a base URL and model name.

Zero-fail fallback

On API timeout or 429, a deterministic local extractor keeps returning contract-valid plans.

Contract-perfect output

Gate G5 URL scrubbing and Blocks A1 / A2 / A5 enforced in code, not left to the LLM.

Checklist- Updated on Public GitHub

- Working prototype code — public or shared GitHub repo - Y
- README with reproducible setup instructions - Y
- Demo video, max 5 minutes (YouTube or Drive link) - Y
- Presentation file (PPT or PDF) - Y

Thank you

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India